

WORK SHEET - 02

KINEMATICS

HINTS & SOLUTIONS
SECTION-A

Q.1 (C)

Let a be the acceleration
at $t = 2n$

$$v_0 = 0 + a \times 2n = 2na \quad \dots(1)$$

at $t = n$

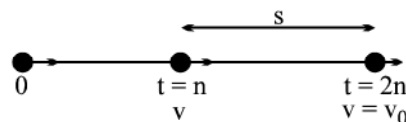
$$v = an \quad \dots(2)$$

between $t = n$ & $t = 2n$

$$v_0^2 - v^2 = 2as \Rightarrow 4n^2a^2 - a^2n^2 = 2as \Rightarrow s = 3n^2a/2$$

From eqⁿ(1) $a = v_0/(2n)$

$$\text{So, } s = \frac{3n^2}{2} \left(\frac{v_0}{2n} \right) = \frac{3v_0n}{4}$$



Q.12 (C)

$$\vec{v}_1 = u_1\hat{i} - gt\hat{j}, \quad \vec{v}_2 = -u_2\hat{i} - gt\hat{j}$$

$$\vec{v}_1 \cdot \vec{v}_2 = 0 \Rightarrow t = \frac{\sqrt{u_1 u_2}}{g}$$

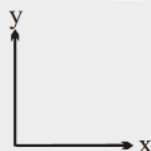
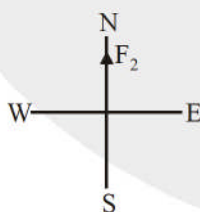
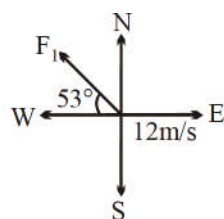
Q.16 (B)

$$s_y u_y t + \frac{1}{2} a_y t^2$$

$$= 30 \times 2 - \frac{1}{2} g 2^2 = 60 - 20$$

$$= 40$$

Q.17 (AC)



Vel. of air = $v\hat{j}$

$$\vec{v}_{AB} = v\hat{j} - 12\hat{i}$$

$$\tan 53^\circ = \frac{v}{12} = \frac{4}{3}$$

$$v = \frac{4 \times 12}{3} = 16 \text{ m/s}$$

$$v_{AB} = \sqrt{(16)^2 + (12)^2} = 20 \text{ m/s}$$

Q.18 (BC)

$$v = 0 - 9 + t^2$$

$$a = 0 - 0 + 2t$$

Q.19 (AD)

Condition for collision in mid air.

$\alpha_{AB} = 0$ and $\vec{v}_{AB}$ should be directed from A to B.

$$\therefore \vec{v}_{AB} = \vec{v}_A - \vec{v}_B$$

$$= 20\hat{i} - [-20\cos(\theta + 30^\circ)\hat{i} + 20\sin(\theta + 30^\circ)\hat{j}]$$

$$= [20 + 20\cos(\theta + 30^\circ)]\hat{i} - 20\sin(\theta + 30^\circ)\hat{j}$$

$$\tan 30^\circ = \frac{h_y}{h_x} = \frac{20\sin(\theta + 30^\circ)}{20 + 20\cos(\theta + 30^\circ)}$$

$$1 + \cos(\theta + 30^\circ) = \sqrt{3} \sin(\theta + 30^\circ)$$

$$1 = \sqrt{3} \sin(\theta + 30^\circ) - \cos(\theta + 30^\circ)$$

$$\frac{1}{2} = \frac{\sqrt{3}}{2} \sin(\theta + 30^\circ) - \frac{1}{2} \cos(\theta + 30^\circ)$$

$$\frac{1}{2} = \sin(\theta + 30^\circ - 30^\circ)$$

$$\sin(\pi/6) = \sin \theta$$

$$\theta = \frac{\pi}{6} = 30^\circ$$

$$\therefore \vec{v}_{AB} = (20 + 20\cos 60^\circ)\hat{i} - 20\sin 60^\circ\hat{j}$$

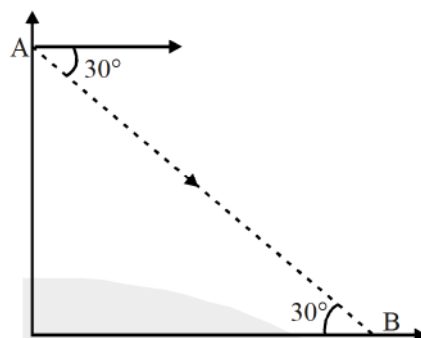
$$= 30\hat{i} - 10\sqrt{3}\hat{j}$$

$$|\vec{v}_{AB}| = \sqrt{(30)^2 + (10\sqrt{3})^2} = 10\sqrt{9+3} = 20\sqrt{3} \text{ ms}^{-1}$$

$$\text{time to collide} = \frac{\text{relative distance}}{\text{relative speed}}$$

$$\text{time to collide} = \frac{200}{20\sqrt{3}} = \frac{10}{\sqrt{3}}$$

Hence answer is (A) and (D)



Q.21 (AB)

Distance travelled by 2nd

particle in 2 sec = $0.5 \times 2 = 1$ m

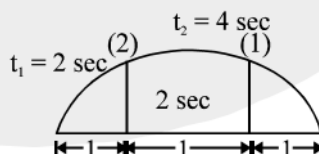
H.Range = $1 + 1 + 1 = 3$ m

Flight time = $4 + 2 = 6$ sec

$$6 = \frac{2u \sin \theta}{g} \quad u \sin \theta = 30$$

$$H = \frac{u^2 \sin^2 \theta}{2g} = \frac{500}{2 \times 10} = 45$$

Particle will strike the ground after 2 sec.



Q.22 (AC)

X

$$u_x = u$$

$$a_x = -g \cos \alpha$$

$$v_x = 0$$

$$v_x = u_x + a_x t$$

$$\therefore t = \frac{u}{g \cos \alpha} = \frac{u}{g \sin \beta}$$

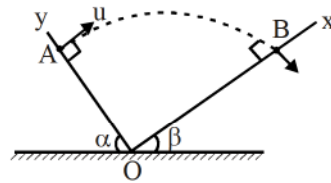
$$0 - u^2 = 2(-g \cos \alpha) s_x$$

$$\therefore s_x = \frac{u^2}{2g \cos \alpha} = \frac{u^2}{2g \sin \beta}$$

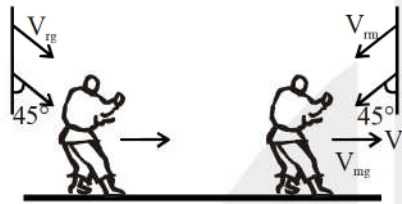
Y

$$u_y = 0$$

$$a_y = -g \sin \alpha$$



Q.23 (CD)



$$\vec{V}_{rg} = \vec{V}_{rm} + \vec{V}_{mg}$$

$$\vec{V}_{rm} = \vec{V}_{rg} - \vec{V}_{mg}$$

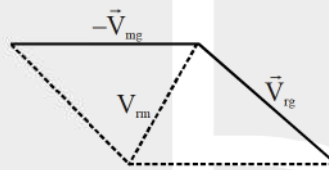
$$V_{rm} \cos 45^\circ = V_{rg} \cos 45^\circ$$

$$V_{rm} = 2\sqrt{2} \text{ m/s} = V_{rg}$$

$$V_{rm} \cos 45^\circ = V_{mg} - V_{rg} \cos 45^\circ$$

$$V_{mg} = 2\sqrt{2} \frac{1}{\sqrt{2}} + 2\sqrt{2} \frac{1}{\sqrt{2}} = 4 \text{ m/s}$$

using $v^2 = u^2 + 2as$ for the motion man,
 $s = 16 \text{ m}$.



SECTION-B

Q.4 [5]

In the frame of train along y-axis

$$0 = 10 \frac{\sqrt{3}}{2} - 10 t$$

$$t = \frac{\sqrt{3}}{2} \quad \therefore T = \sqrt{3} \text{ sec.}$$

Along x-axis

$$1.15 = \left(10 \frac{1}{2}\right) (\sqrt{3}) \frac{1}{2} a (\sqrt{3})^2$$

$$\therefore a = 5 \text{ m/s}^2$$

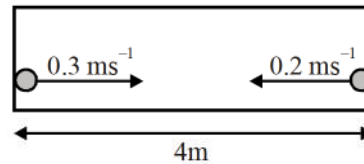
Q.5

[8]

$$V_{\text{rel}} = 0.5 \text{ m/s}$$

$$a_{\text{rel}} = 0$$

$$t = \frac{S_{\text{rel}}}{V_{\text{rel}}} = \frac{4}{0.5} = 8 \text{ sec.}$$



Q.6

[5]

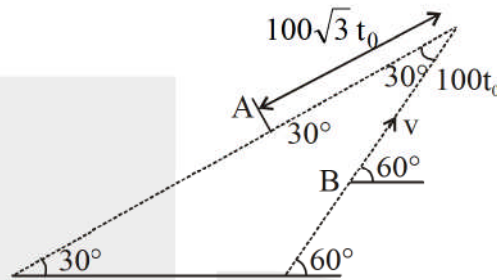
$$V_{BA} = \vec{V}_B - \vec{V}_A$$

$$\Rightarrow v \cos 30^\circ = 100\sqrt{3}$$

$$\Rightarrow v = 200 \text{ m/s}$$

$$V_{BA} = v \sin 30^\circ = 100 \text{ m/s}$$

$$t = S_{BA}/V_{BA} = 5 \text{ sec.}$$



Q.7

[2.00]

$$|\vec{A} + \vec{B}| = 2a \cos \frac{\omega t}{2}$$

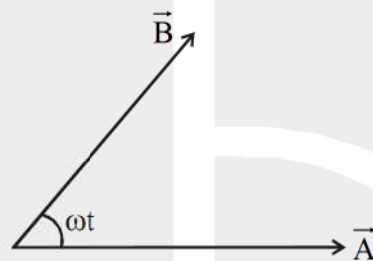
$$|\vec{A} - \vec{B}| = 2a \sin \frac{\omega t}{2}$$

from given conditions.

$$2a \frac{\cos \omega t}{2} = \sqrt{3} \cdot 2a \sin \frac{\omega t}{2}$$

$$\frac{\omega t}{2} = \frac{\pi}{6} \Rightarrow \omega t = \frac{\pi}{3}$$

$$\Rightarrow \frac{\pi}{6} \times t = \frac{\pi}{3} \Rightarrow t = 2.00 \text{ sec.}$$



Q.8

[6.30]

$$\frac{dx}{dt} = v_0 e^{-t/\tau}$$

$$\int_0^x dx = \int_0^{\tau=4} v_0 e^{-t/\tau} dt$$

$$x = v_0 \tau (1 - e^{-1})$$

$$= v_0 \times 4 \times 0.63 \dots (1)$$

$$I = mv_{(0)}$$

$$1 = 0.4 \times v_0$$

$$v_0 = 5/2$$

Put in (1)

$$x = \frac{5}{2} \times 4 \times 0.63 = 6.30$$

Q.9

[4]

Sol. Average velocity = $\frac{\text{Total displacement}}{\text{Total time}}$

$$\text{Total time taken} = t_1 + t_2 + t_3 + \dots$$



$$= t_1 + \frac{t_1}{\alpha} + \frac{t_1}{\alpha^2} + \dots$$

$$\text{Total time} = \frac{t_1}{1 - \frac{1}{\alpha}}$$

$$\text{Total displacement} = v_1 t_1 + v_2 t_2 + \dots$$

$$= v_1 t_1 + \frac{v_1}{\alpha} \cdot \frac{t_1}{\alpha} + \dots$$

$$= \frac{v_1 t_1}{1 - \frac{1}{\alpha^2}}$$

On solving

$$\langle v \rangle = \frac{v_1 \alpha}{\alpha + 1} = 0.8 v_1$$

$$\alpha = 4.00$$

Q.11 [0.95]

$$d = \frac{v^2 \sin 2\theta}{g}$$

$$H_{\max} = \frac{v^2 \sin^2 \theta}{2g}; \frac{1}{2} g_{\text{eff}} t^2 = H_{\max}$$

$$\text{New range} = \frac{1}{2} + d_1 \Rightarrow t^2 = \frac{2H_{\max}}{g_{\text{eff}}}$$

$$= \frac{2 \times v^2 \sin^2 \theta}{2g \left(\frac{g}{0.81} \right)}$$

$$d_1 = v \cos \theta \cdot t$$

$$= \frac{v^2 \sin^2 \theta \cos \theta \times 0.9}{g}; t = \sqrt{\frac{v^2 \sin^2 \theta \times 0.81}{g^2}}; t = \frac{0.9 v \sin \theta}{g}$$

$$d' = \frac{v^2 \sin 2\theta}{2g} + \frac{v^2 \sin 2\theta \times 0.9}{2g}$$

$$= \frac{v^2 \sin 2\theta}{g} \left(\frac{1.0}{2} \right) = 0.95d$$

$$n = 0.95$$

